

MAKE YOUR OWN DIGITAL ALARM CLOCK



■ ATISH GUPTA

Time management is very important in today's fast-paced life. An alarm clock helps you manage your time effectively by alerting you of appointments or other important tasks.

Here we have described a microcontroller-based digital alarm clock. The time and alarm can be customised by the user, and are shown on the liquid crystal display (LCD) of the system.

Circuit description

Fig. 1 shows the circuit of the microcontroller-based digital alarm clock. It comprises microcontroller AT89C51, an LCD module, regulator 7805 and a few discrete components.

Microcontroller AT89C51 is a low-

power, high-performance CMOS 8-bit microcomputer with 4 kB of Flash programmable and erasable read-only memory (PEROM). It has the following standard features: 4 kB of Flash, 128 bytes of RAM, 32 I/O lines, two 16-bit timers/counters, five-vector two-level interrupt architecture, a full-duplex serial port, and on-chip oscillator and clock circuitry. In addition, the AT89C51 is designed with static logic for operation down to zero frequency and supports two software-selectable power-saving modes. The idle mode stops the CPU while allowing the RAM, timers/counters, serial port and interrupt system to continue functioning. The power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next hardware reset.

The system clock also plays a

significant role in operation of the microcontroller. An 11.0592MHz quartz crystal is connected to pins 18 and 19, to provide basic clock to the microcontroller. Power-on reset is provided by electrolytic capacitor C3 and resistor R1. Switch S1 is used for manual reset. The clock time is shown on the LCD.

Port pins P1.0 through P1.7 of the microcontroller are connected to data pins D0 through D7 of the LCD, respectively. Port pins P3.7, P3.6 and P3.5 are connected to register-select RS, read-write R/W and enable E of the LCD, respectively. Port pins P1.0 through P1.7 are pulled high by resistor network RNW1, while port pins P3.5 through P3.7 are pulled high by resistors R6 through R8, respectively.

All the data is sent to the LCD in

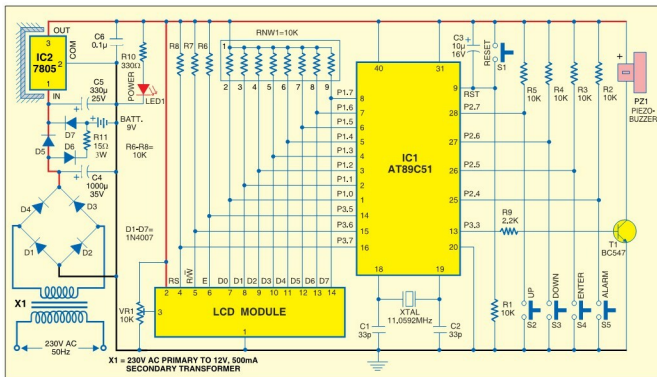


Fig. 1: Circuit of microcontroller-based digital alarm clock